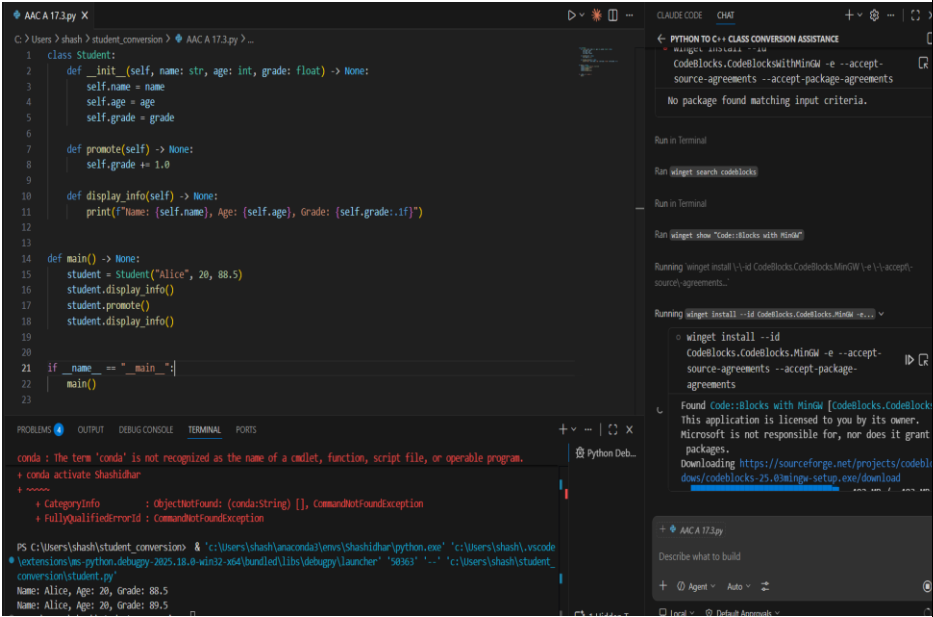


Name:A.Shashidhar H.No:2303A51798 Batch:26

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
Program Name: B. Tech		Assignment Type: Lab	Academic Year:2025-2026
Course Coordinator Name		Dr. Rishabh Mittal	
Instructor(s) Name		Mr. S Naresh Kumar	
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		Dr. Sasanko Shekhar Gantayat	
		Mr. Md Sallauddin	
		Dr. Mathivanan	
		Mr. Y Srikanth	
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		Dr. R. Prashant Kumar	
		Mr. Ankushavali MD	
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		Mr. Bingi Raju	
CourseCode	23CS002PC304	Course Title	AI Assisted Coding
Year/Sem	III/II	Regulation	R23
Date and Day of Assignment	Week10 – Wednesday	Time(s)	23CSBTB01 To 23CSBTB52
Duration	2 Hours	Applicable to Batches	All batches
Assignment Number:19.3(Present assignment number)/24(Total number of assignments)			

Q.No.	Question	Expected Time to complete
1	Lab 19 – Code Translation: Converting Between Programming Languages Lab Objectives:	Week10 – Wednesday

	• Understand how AI tools can assist in translating code between different programming languages. • Learn to verify correctness and functionality after translation. • Explore syntactic and semantic differences between languages (e.g., Python, Java, C++). • Practice debugging and optimizing AI-translated code.	
	Task 1 – Python to C++ Conversion Task: Translate a simple Python class into C++ using AI assistance. Instructions: • Prompt AI to convert a Python class representing a Student with attributes and methods into equivalent C++ code • Ensure proper handling of: ○ Constructors ○ Data types ○ Access specifiers • Compile and run both versions to verify output consistency Expected Output: An equivalent working C++ class translated from Python.  Task 2 – Java to Python Function Conversion Task: Convert a Java method that checks whether a number is prime into Python. Instructions: • Ask AI to translate a Java isPrime() method into Python • Test the function with multiple inputs	

- Ensure the Python version follows Pythonic syntax and logic

Expected Output:

A correct Python function equivalent to the original Java prime-checking method.

```

1 import math
2
3 def is_prime(n: int) -> bool:
4     if n <= 1:
5         return False
6     if n == 2:
7         return True
8     if n % 2 == 0:
9         return False
10
11     limit = int(math.isqrt(n))
12     for i in range(3, limit + 1, 2):
13         if n % i == 0:
14             return False
15     return True
16
17
18 # Test with multiple inputs
19 test_values = [-5, 0, 1, 2, 3, 4, 5, 9, 11, 25, 29, 97, 100, 101]
20
21 for value in test_values:
22     print(f"{value}: {is_prime(value)}")

```

Task
Convert a Java method that checks whether a number is prime into Python.
Instructions:
• Ask AI to translate a Java isPrime() method into Python
• Test the function with multiple inputs
• Ensure the Python version follows Pythonic syntax and logic
Expected Output:
A correct Python function equivalent to the original Java prime-checking method.

Python code (Pythonic conversion of Java-style isPrime, with multiple tests):

```

import math

def is_prime(n: int) -> bool:
    if n <= 1:
        return False
    if n == 2:
        return True
    if n % 2 == 0:
        return False

    limit = int(math.isqrt(n))
    for i in range(3, limit + 1, 2):
        if n % i == 0:
            return False
    return True

```

Terminal Output:

```

PS C:\Users\shash\student_conversion> cd 'c:\Users\shash\student_conversion' & 'c:\Users\shash\anaconda3\envs\shash\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '58613' -- "c:\Users\shash\student_conversion\AAC A 17.3.py"
9: False
11: True
25: False
29: True
97: True
100: False
101: True

```

Task 3 – Pseudocode to Python Implementation

Task:

Translate a given pseudocode algorithm into Python using AI.

Instructions:

- Provide AI with pseudocode for sorting numbers (e.g., bubble sort or selection sort)
- Ask AI to convert it into executable Python code
- Validate correctness using sample input lists

Expected Output:

A working Python program that correctly implements the pseudocode logic.

```

AAC A 17.3.py X
C:\Users\shash> student_conversion > AAC A 17.3.py ...
1 def bubble_sort(nums):
2     arr = nums[:]
3     n = len(arr)
4     for i in range(n):
5         swapped = False
6         for j in range(0, n - i - 1):
7             if arr[j] > arr[j + 1]:
8                 arr[j], arr[j + 1] = arr[j + 1], arr[j]
9                 swapped = True
10        if not swapped:
11            break
12        return arr
13
14
15 def main():
16     test_lists = [
17         [64, 34, 25, 12, 22, 11, 90],
18         [5, 1, 4, 2, 8],
19         [3, 3, 2, 1, 5, 0],
20         [],
21         [10],
22     ]
23
24     for lst in test_lists:
25         print("Input :", lst)
26         print("Sorted:", bubble_sort(lst))
27         print()
28
29 if __name__ == "__main__":
30     main()
31
32

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\shash\student_conversion> c:\cd 'c:\Users\shash\student_conversion'; & "c:\Users\shash\anaconda3\envs\shashidhar\python.exe" "c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher" "56797" "--" "c:\Users\shash\student_conversion\AAC A 17.3.py"

Input : []
Sorted: []

Input : [10]
Sorted: [10]

Python Deb...

def bubble_sort(nums):
arr = nums[:]
n = len(arr)
for i in range(n):
swapped = False
for j in range(0, n - i - 1):
if arr[j] > arr[j + 1]:
arr[j], arr[j + 1] = arr[j + 1], arr[j]
swapped = True
if not swapped:
break
return arr

def main():
test_lists = [
[64, 34, 25, 12, 22, 11, 90],
[5, 1, 4, 2, 8],
[3, 3, 2, 1, 5, 0],
[],
[10],
]

for lst in test_lists:
print("Input :", lst)
print("Sorted:", bubble_sort(lst))
print()

→ AAC A 17.3.py
Describe what to build

🔍 Agent 📄 Data 🔄

```

AAC A 17.3.py X
C:\Users\shash> student_conversion > AAC A 17.3.py > ...
15 def main():
16     test_lists = [
17         [0, 0, 0, 0, 0, 0, 0, 0],
18         [5, 1, 4, 2, 8],
19         [3, 3, 2, 1, 5, 0],
20         [],
21         [10],
22     ]
23
24     for lst in test_lists:
25         print("Input :", lst)
26         print("Sorted:", bubble_sort(lst))
27         print()
28
29
30 if __name__ == "__main__":
31     main()
32

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\shash\student_conversion> c:\cd 'c:\Users\shash\student_conversion'; & "c:\Users\shash\anaconda3\envs\shashidhar\python.exe" "c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher" "56797" "--" "c:\Users\shash\student_conversion\AAC A 17.3.py"

Input : []
Sorted: []

Input : [10]
Sorted: [10]

Task 4 – SQL to Pandas Query

Task:

Translate a **SQL query** into a **Pandas DataFrame operation** in Python.

Instructions:

- Provide AI with a SQL query (e.g., SELECT name, salary FROM employees WHERE salary > 50000).
- Ask AI to generate equivalent Pandas code.
- Test the generated code on a sample DataFrame.

Expected Output:

- Equivalent Pandas query that retrieves the correct subset of data.

The screenshot shows a VS Code editor with a Python file named 'AAC A 17.3.py'. The code defines a DataFrame 'employees' with columns 'name', 'salary', and 'department'. It filters the DataFrame for employees with a salary greater than 50,000 and prints the result. The terminal window at the bottom shows the command to run the script and the output, which lists the names and salaries of the filtered employees. The sidebar on the right contains a task description for 'Task 4 - SQL to Pandas Query' and a 'PYTHON TO C++ CLASS CONVERSION ASSISTANCE' panel.

```
1 import pandas as pd
2
3 employees = pd.DataFrame({
4     "name": ["Alice", "Bob", "Charlie", "Diana", "Evan"],
5     "salary": [48000, 62000, 51000, 45000, 75000],
6     "department": ["HR", "Engineering", "Sales", "Finance", "Engineering"],
7 })
8
9
10
11 result = employees.loc[employees["salary"] > 50000, ["name", "salary"]]
12
13 print(result)
```

Python Deb...

```
name salary
1 Bob 62000
2 Charlie 51000
4 Evan 75000
```

Python Deb...

```
name salary
1 Bob 62000
2 Charlie 51000
4 Evan 75000
```

Task 5 – Real-Time Application: Algorithm Translation Across Languages

Scenario:

A company maintains algorithms written in different programming languages.

Examples:

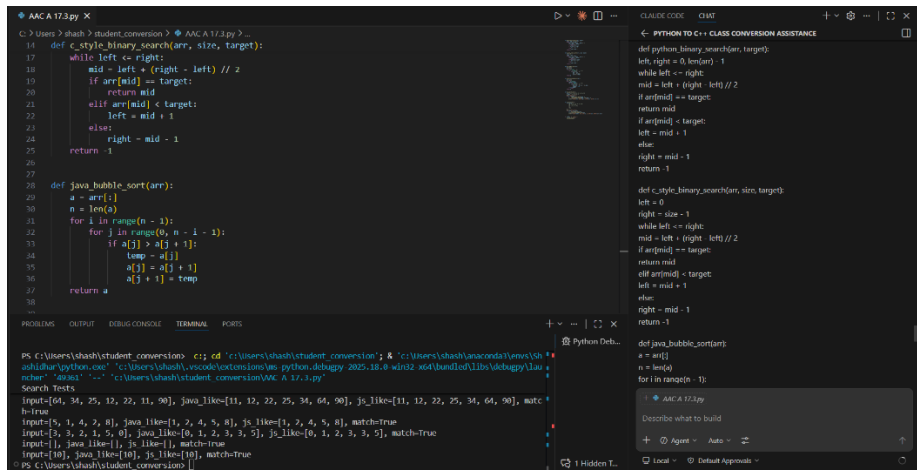
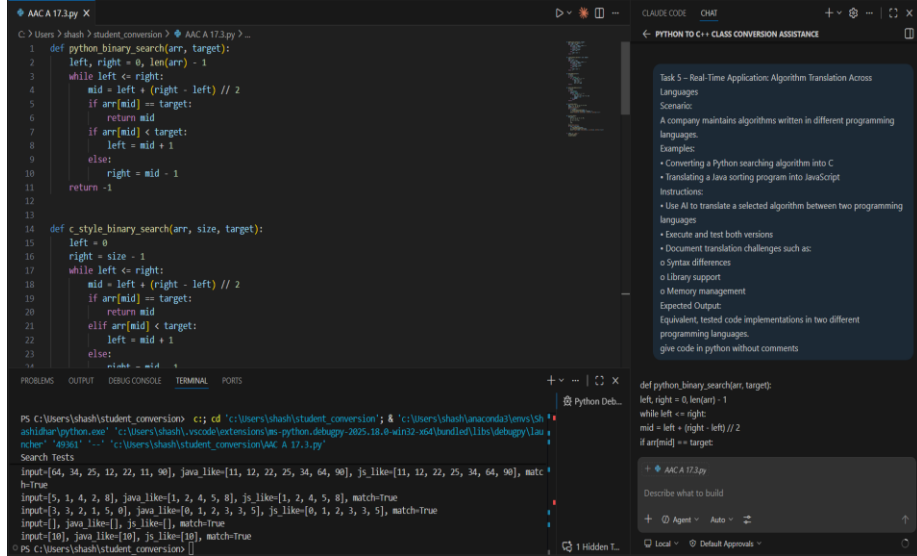
- Converting a Python searching algorithm into C
- Translating a Java sorting program into JavaScript

Instructions:

- Use AI to translate a selected algorithm between two programming languages
- Execute and test both versions
- Document translation challenges such as:
 - Syntax differences
 - Library support
 - Memory management

Expected Output:

Equivalent, tested code implementations in two different programming languages.



```
AAC A 17.3.py X
C: > Users > shash > student_conversion > AAC A 17.3.py > ...

40 def javascript_bubble_sort(arr):
41     a = arr[:]
42     n = len(a)
43     for i in range(n):
44         swapped = False
45         for j in range(0, n - i - 1):
46             if a[j] > a[j + 1]:
47                 a[j], a[j + 1] = a[j + 1], a[j]
48                 swapped = True
49         if not swapped:
50             break
51     return a
52
53
54 def run_search_tests():
55     data = [3, 7, 12, 19, 24, 31, 45, 58]
56     targets = [3, 24, 58, 9]
57     print("Search Tests")
58     for t in targets:
59         p = python_binary_search(data, t)
60         c = c_style_binary_search(data, len(data), t)
61         print(f"target={t}, python={p}, c_style={c}, match={p == c}")
62
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\student_conversion> c++; cd 'c:\Users\shash\student_conversion'; & 'c:\Users\shash\anaconda3\envs\Sh
ashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\lau
ncher' '49361' '--' 'c:\Users\shash\student_conversion\AAC A 17.3.py'
Search Tests
input=[64, 34, 25, 12, 22, 11, 90], java_like=[11, 12, 22, 25, 34, 64, 90], js_like=[11, 12, 22, 25, 34, 64, 90], matc
h=True
input=[5, 1, 4, 2, 8], java_like=[1, 2, 4, 5, 8], js_like=[1, 2, 4, 5, 8], match=True
input=[3, 3, 2, 1, 5, 0], java_like=[0, 1, 2, 3, 3, 5], js_like=[0, 1, 2, 3, 3, 5], match=True
input=[], java_like=[], js_like=[], match=True
input=[10], java_like=[10], js_like=[10], match=True
```

```
AAC A 17.3.py X
C: > Users > shash > student_conversion > AAC A 17.3.py > ...

63
64 def run_sort_tests():
65     test_cases = [
66         [64, 34, 25, 12, 22, 11, 90],
67         [5, 1, 4, 2, 8],
68         [3, 3, 2, 1, 5, 0],
69         [],
70         [10],
71     ]
72     print("\nSort Tests")
73     for case in test_cases:
74         j = java_bubble_sort(case)
75         js = javascript_bubble_sort(case)
76         print(f"input={case}, java_like={j}, js_like={js}, match={j == js}")
77
78
79 if __name__ == "__main__":
80     run_search_tests()
81     run_sort_tests()
82
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\student_conversion> c++; cd 'c:\Users\shash\student_conversion'; & 'c:\Users\shash\anaconda3\envs\Sh
ashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\lau
ncher' '49361' '--' 'c:\Users\shash\student_conversion\AAC A 17.3.py'
Search Tests
input=[64, 34, 25, 12, 22, 11, 90], java_like=[11, 12, 22, 25, 34, 64, 90], js_like=[11, 12, 22, 25, 34, 64, 90], r
h=True
input=[5, 1, 4, 2, 8], java_like=[1, 2, 4, 5, 8], js_like=[1, 2, 4, 5, 8], match=True
input=[3, 3, 2, 1, 5, 0], java_like=[0, 1, 2, 3, 3, 5], js_like=[0, 1, 2, 3, 3, 5], match=True
input=[], java_like=[], js_like=[], match=True
input=[10], java_like=[10], js_like=[10], match=True
```

Note: Report should be submitted as a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots.